Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 7

Title: Developing Finance KPI Dashboards

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Problem Statement

You are a Financial Analyst in a manufacturing company. Senior management wants to monitor budget performance, actual expenses, budget utilization, and spending variations across departments, regions, categories, and payment methods. The objective is to create an interactive Finance KPI Dashboard in Power BI that helps management evaluate financial performance, identify overspending areas, compare budget with actual amounts, and support better financial planning and decision-making.

Methodology / Steps Followed

Step 1: Data Collection

- Collected the financial budget and actual transaction dataset.
- The dataset contains financial information such as year, quarter, month, department, category, region, budget amount, actual amount, variance, and payment method.
- The dataset was selected to analyze budget utilization, expenses, and financial performance.

Step 2: Data Import

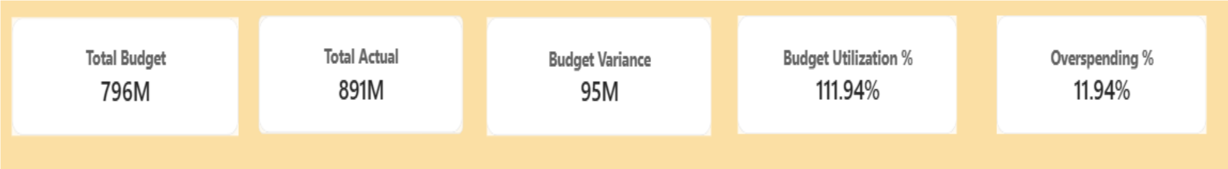
- Opened Power BI Desktop.
- Selected Get Data → Excel/appropriate data source.
- Imported the financial dataset into Power BI.
- Verified that the required financial columns were loaded correctly.

Step 3: Data Cleaning and Transformation

Using Power Query:

- Checked for missing and null values.
- Checked for duplicate records.
- Verified data types of all columns.
- Correctly formatted Year, Quarter, Month, Department, Category, Region, and Payment Method.
- Verified Budget Amount and Actual Amount as numerical fields.
- Created/verified the required variance and utilization calculations.
- Checked the final dataset before creating visualizations.

KPI Cards Analysis



1. Total Budget

Value: 796M

- Represents the total budget allocated for the selected period.
- Helps management understand the planned financial expenditure.
- Provides a baseline for comparing actual spending.

2. Total Actual

Value: 891M

- Represents the total actual amount spent.
- Helps management monitor actual financial expenditure.
- Comparing actual expenditure with the budget helps identify overspending.

3. Budget Variance

Value: 95M

- Represents the difference between the actual amount and the budget amount.
- A positive variance in this dashboard indicates that actual spending is higher than the budget.
- Helps management identify areas requiring financial control.

4. Budget Utilization %

Value: 111.94%

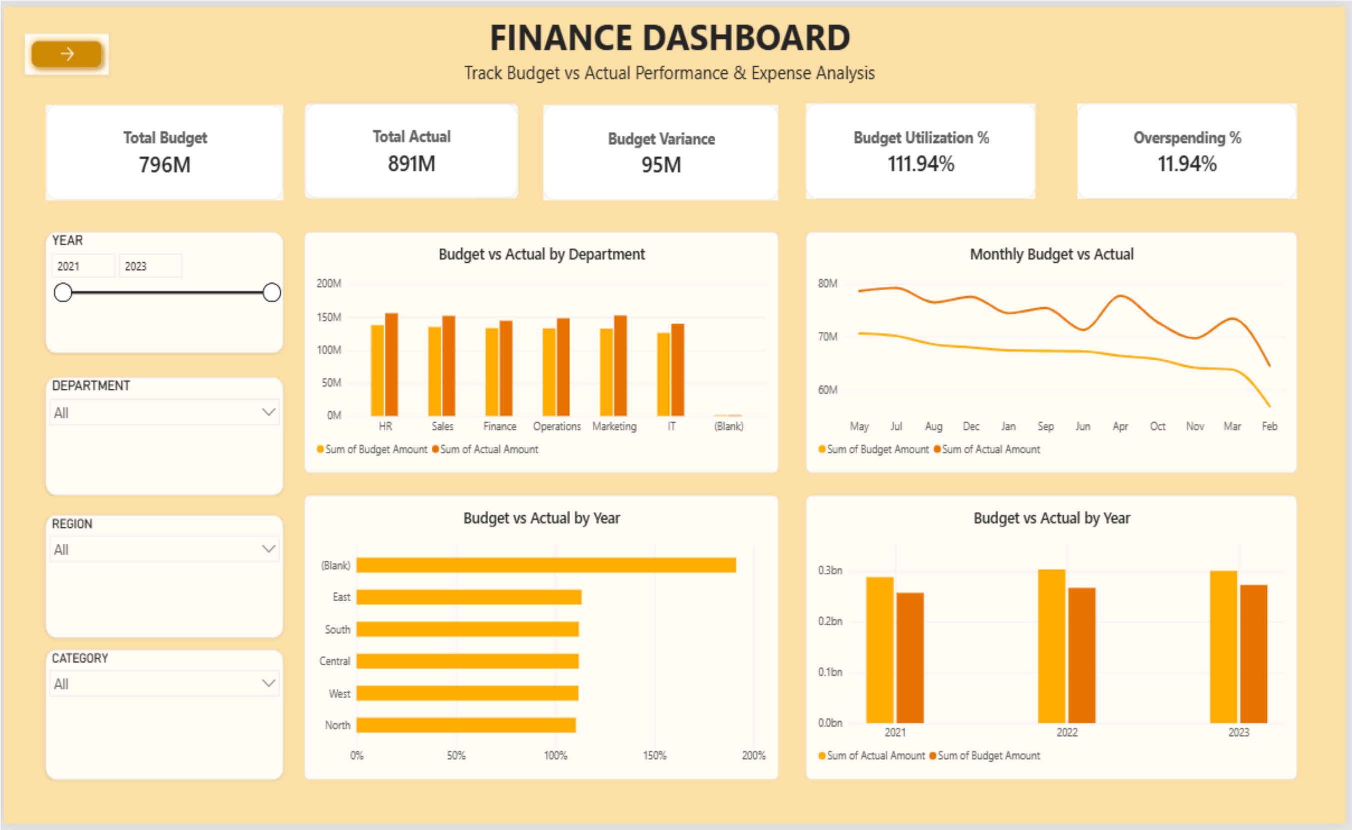
- Represents actual spending as a percentage of the allocated budget.
- A value above 100% indicates that the organization has spent more than the planned budget.
- Helps management monitor budget consumption.

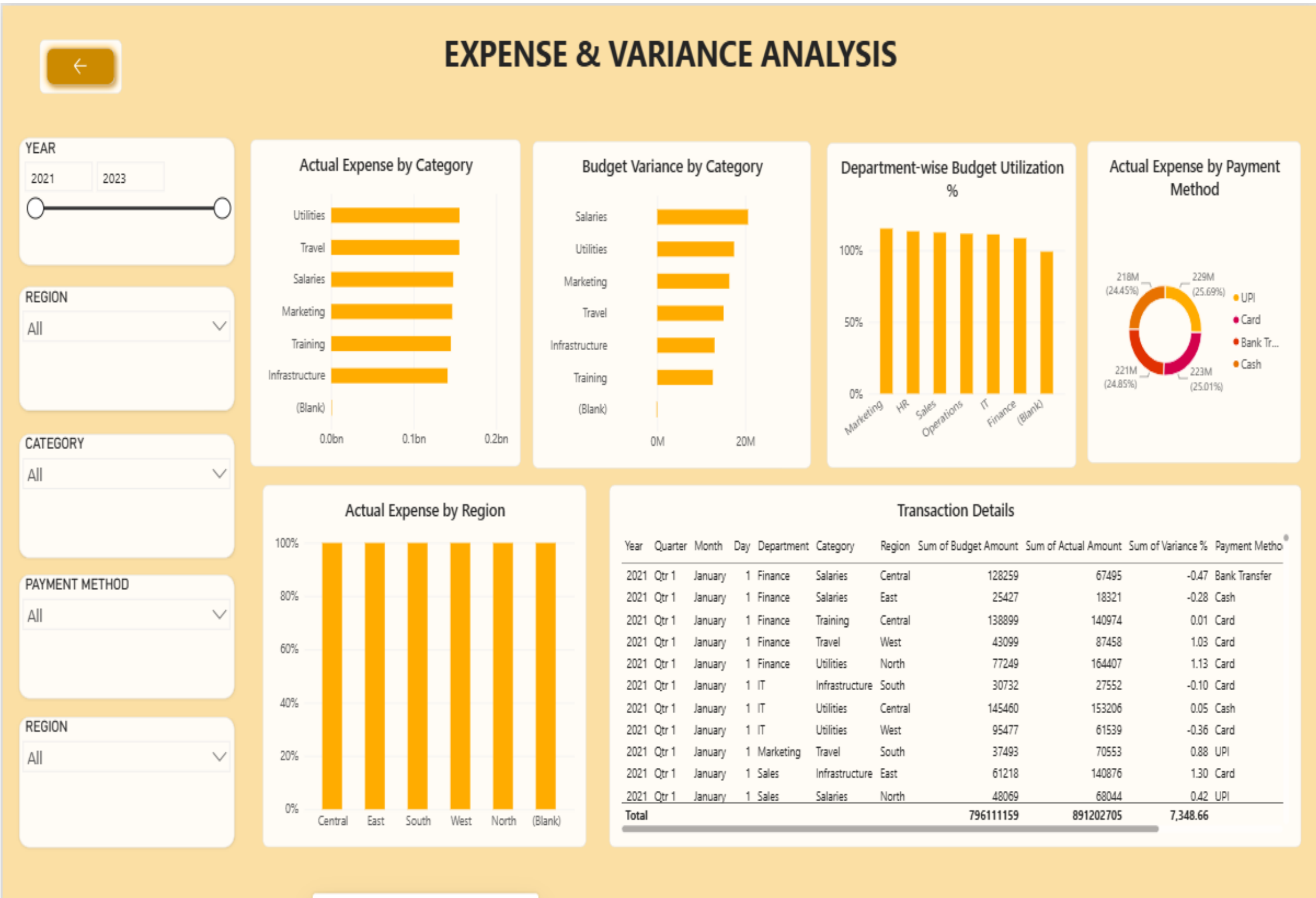
5. Overspending %

Value: 11.94%

- Represents the percentage by which actual spending exceeds the allocated budget.
- Helps management quickly identify the overall level of overspending.
- Higher overspending indicates the need for closer expense monitoring.

Visualizations Analysis





1. Budget vs Actual by Department — Clustered Column Chart

Purpose:
Shows the comparison between budgeted and actual amounts across departments.

Observation:

- The chart compares financial performance across HR, Sales, Finance, Operations, Marketing, and IT.
- Actual amounts are generally higher than the corresponding budget amounts.
- The comparison helps identify departments where spending exceeds planned levels.

Business Insight:

- Management can identify departments with higher spending.
- Departments with significant differences between budget and actual expenditure can be investigated.
- Department-level budget controls can be introduced where necessary.

2. Monthly Budget vs Actual — Line Chart

Purpose:
Shows monthly trends in budgeted and actual financial amounts.

Observation:

- The visualization shows changes in budget and actual amounts across months.
- The two lines allow management to compare planned expenditure with actual expenditure over time.
- Some months show a larger gap between budget and actual values.

Business Insight:

- Monthly trends help identify periods of higher expenditure.
- Management can investigate unusual increases in actual spending.
- The analysis can support future monthly budget planning.

3. Budget vs Actual by Region — Bar Chart

Purpose:
Compares budget utilization/performance across different regions.

Observation:

- The visualization compares financial performance across Central, East, South, West, North, and other available region categories.
- Regional differences can be observed in the level of budget utilization.

Business Insight:

- Management can identify regions with comparatively higher budget utilization.
- Regional spending patterns can be investigated to improve cost control.
- Financial resources can be planned according to regional requirements.

4. Budget vs Actual by Year — Column Chart

Purpose:
Compares annual budget and actual amounts.

Observation:

- The dashboard compares financial amounts for 2021, 2022, and 2023.
- Actual spending is higher than budgeted spending across the displayed years.

Business Insight:

- The visualization helps management evaluate year-to-year financial performance.
- Persistent differences between budget and actual expenditure indicate the need for improved budget forecasting.
- Annual trends can support future financial planning.

5. Actual Expense by Category — Bar Chart

Purpose:
Shows actual expenses across different expense categories.

Observation:

- Categories such as Utilities, Travel, Salaries, Marketing, Training, and Infrastructure are displayed.
- Utilities and Travel represent relatively high actual expenses in the dashboard.

Business Insight:

- Management can identify major expense categories.
- High-expense categories can be reviewed for cost-saving opportunities.
- Category-level analysis supports better expense management.

6. Budget Variance by Category — Bar Chart

Purpose:
Shows the difference between budgeted and actual spending across expense categories.

Observation:

- Salaries show a comparatively high budget variance.
- Utilities, Marketing, Travel, Infrastructure, and Training also contribute to the overall variance.

Business Insight:

- Categories with high variance require closer financial monitoring.
- Management can investigate the reasons for deviations from the planned budget.
- Variance analysis can improve future budget estimation.

7. Department-wise Budget Utilization % — Column Chart

Purpose:
Shows the percentage of budget utilized by each department.

Observation:

- Most departments show budget utilization close to or above 100%.

- Several departments exceed the planned budget.

Business Insight:

- Departments exceeding 100% utilization should be reviewed.
- This visualization helps management identify departments requiring stronger spending controls.
- Department-level utilization can be used for future budget allocation.

8. Actual Expense by Payment Method — Donut Chart

Purpose:
Shows the distribution of actual expenses according to payment method.

Observation:

- The dashboard displays payment methods such as UPI, Card, Bank Transfer, and Cash.
- The expense distribution is relatively spread across the different payment methods.

Business Insight:

- Management can understand how financial transactions are distributed across payment channels.
- Payment-method analysis can support transaction monitoring and financial control.
- It can also help identify the most frequently used payment channels.

9. Actual Expense by Region — Column Chart

Purpose:
Analyzes actual expenses across different regions.

Observation:

- Actual expense distribution is displayed for Central, East, South, West, North, and other available regions.
- The visualization allows comparison of regional expenditure.

Business Insight**:**

- Management can identify regions with higher expenditure.
- Regional expense patterns can be compared with business requirements and allocated budgets.
- High-spending regions can be investigated for possible cost optimization.

10. Transaction Details — Table

Purpose:
Provides detailed transaction-level financial information.

Displayed information includes:

- Year
- Quarter
- Month
- Day

- Department
- Category
- Region
- Budget Amount
- Actual Amount
- Variance %
- Payment Method

Business Insight:

- The table allows management to drill down from summary-level KPIs to individual transactions.
- It helps identify specific transactions contributing to high expenses or budget variance.
- Detailed transaction analysis supports financial auditing and investigation.

Overall Finance Insights

- The total budget is **796M**, while total actual expenditure is **891M**.
- Actual expenditure exceeds the allocated budget by approximately **95M**.
- Overall budget utilization is **111.94%**, indicating that spending has exceeded the planned budget.
- Overall overspending is **11.94%**.
- Department-wise analysis helps identify departments with higher actual expenditure compared with their budgets.
- Monthly analysis helps identify changes and fluctuations in financial expenditure.
- Category analysis shows that expenses such as **Utilities, Travel, Salaries, Marketing, Training, and Infrastructure** contribute to overall spending.
- Budget variance analysis helps identify categories where actual spending differs significantly from planned spending.
- Regional analysis provides insights into differences in financial expenditure across regions.
- Payment-method analysis helps management understand the distribution of transactions across UPI, Card, Bank Transfer, and Cash.
- Transaction-level details allow management to investigate individual financial records.
- The dashboard can support better **budget control, expense monitoring, variance analysis, and financial planning**.

Key Management Findings

1. Overall Overspending

Actual expenditure of **891M** is higher than the budget of **796M**, resulting in approximately **95M** higher spending.

2. Budget Utilization

Budget utilization of **111.94%** indicates that the organization has exceeded its planned expenditure level.

3. Expense Control

Major expense categories should be monitored regularly to reduce unnecessary expenditure.

4. Department Monitoring

Departments with utilization above 100% should be reviewed to understand the reasons for overspending.

5. Regional Monitoring

Regional expenditure comparison can help management identify locations with comparatively higher financial requirements.

6. Variance Monitoring

Budget variance analysis can help management take corrective action before financial deviations become significant.

Conclusion

The Finance KPI Dashboard provides an interactive view of the organization's financial performance by comparing budgeted and actual expenditure. The dashboard highlights important KPIs such as Total Budget, Total Actual, Budget Variance, Budget Utilization, and Overspending. It also provides department-wise, category-wise, region-wise, monthly, yearly, and payment-method analysis.

The analysis shows that actual expenditure of 891M exceeds the total budget of 796M, resulting in approximately 95M variance and 111.94% budget utilization. The dashboard therefore helps management identify overspending areas, monitor expenses, investigate budget deviations, and improve financial planning.

Overall, the Finance KPI Dashboard supports effective budget management, expense control, financial monitoring, and data-driven decision-making.

GITHUBLINK: https://github.com/lohithslohiths110405-ui/Data-Analytics-for-Business/tree/main/Week_7_EXP